Predictive power of large language models in a pragmatically-driven sentence processing task

Background Although the relationship between reading time and surprisal has been established as linear [e.g., 1-4], Huang et al. (2024) show that the effect of surprisal alone is not enough to explain the processing difficulty in resolving syntactic ambiguity, including the case of relative clause (RC) attachment [5]. In addition, recent studies have shown that large language models (LLMs) trained with more parameters and data are worse at predicting reading time despite achieving lower perplexities [7-8]. Yet these same models tend to perform better in pragmatic tasks [8], and have been shown to leverage pragmatic inferences to resolve attachment for ambiguous RCs [10]. Specifically, like humans [11], LLMs shift their bias toward high attachment when an object-biased implicit causality (IC) verb (*detest* in (1b)) is used compared to a non-IC verb (*babysits* in (1a)), here leading to a higher probability assigned to the plural auxiliary "*are*". In this study, we examine whether the negative relationship between perplexity and psychometric predictive power holds when enhanced pragmatic abilities have been shown to lead to better syntactic and, ultimately, next-word predictions. Specifically, if the negative relationship between perplexity and psychometric predictive power holds in larger models, then larger models would be worse at predicting human reading time than smaller models. Yet, if the larger number of parameters results in enhanced pragmatic inference abilities, and these abilities are leveraged to make better syntactic predictions, then larger models would perform better at predicting reading time.

Methods We evaluated five models: GPT-2, Llama-3.2-1B, Llama-3.2-3B, and the instruction-tuned versions of the two Llama models. We obtained the surprisal of words in each region of the 40 sentences used in the self-paced reading task in [10] (example in (2)). We used generalized additive models (GAM) to predict the average reading time across participants in [11] from surprisal and two control variables, word length and frequency. We compared the full GAM models with a baseline GAM model that included only the control variables and calculated the change in log likelihood (ΔLogLik) to measure the effect of surprisal. Additionally, we fit a GAM model to the measures in the non-critical regions and evaluated how well the model predicted the reading time of the critical regions, including the auxiliary verb and the two subsequent spillover regions. We used root mean squared error (RMSE) to measure the difference between the predicted reading time and the actual reading time in the critical regions.

Results The relationship between perplexity and ΔLogLik is negative (Fig. 1), suggesting that the better the model predicts the next word (i.e., lower perplexity), the better it simulates reading time (i.e., larger ΔLogLik). With respect to the generalizability of the GAM models fit to non-critical regions to reading time in the critical regions, there is a positive relationship between perplexity and RMSE values (Fig. 2): GPT-2 had the largest RMSE, followed by the two instruction-tuned Llama models and then the two base models. The smaller Llama models had lower RMSE values, with Llama-3.2-1B lowest, suggesting better generalization of the relationship between surprisal and reading time from non-critical regions to critical regions.

Conclusion Across the five attested models, instruction-tuned models tend to have lower perplexity and higher ΔLogLik than smaller models, but are worse at generalizing the relationship between surprisal and reading time in non-critical regions. In addition, for models such as GPT-2, which reportedly show limited pragmatic inference abilities and fail to use them in processing RCs, the relationship between surprisal and human reading time in non-critical regions did not generalize well to critical regions. Thus, larger models with better pragmatic inference abilities might not always simulate human reading time better.

Example sentences

(1) a. Melissa babysits the children of the musician who is/are
b. Melissa detests the children of the musician who is/are

(2) Melissa babysits/detests the children of the musician who is/are generally arrogant and rude.

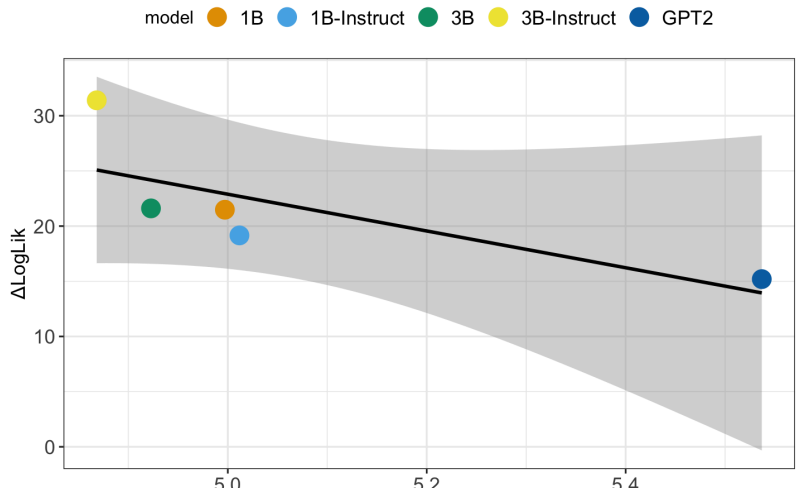

Figure 1.The ΔLogLik against model perplexity for each model

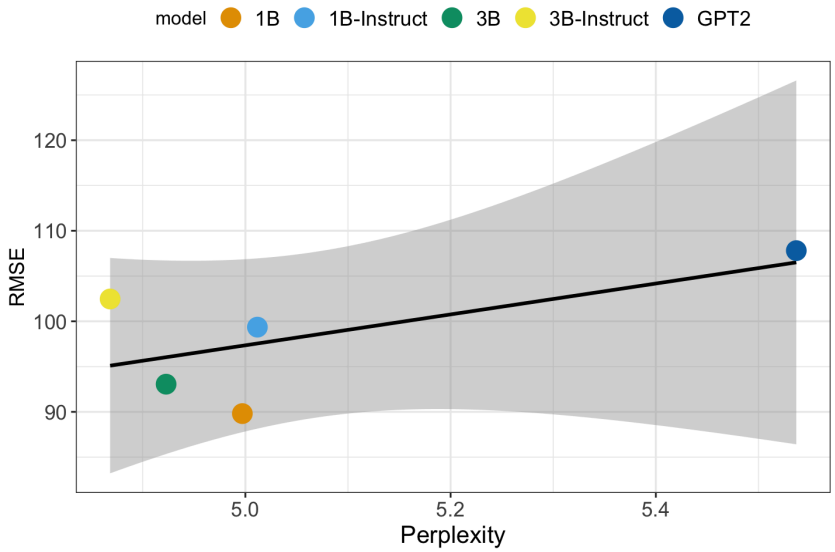

Figure 2.The RMSE against model perplexity for each model.

References

[1] Smith & Levy (2013). *Cognition*. [2] Fossum & Levy (2012). *CMCL 2012*. [3] Goodkind & Bicknell (2018). *CMCL 2018*. [4] Wilcox et al. (2020). *Proceedings of Cogsci.* [5] Huang et al. (2024). *Journal of Memory and Language.* [6] Oh & Schuler (2023). *TACL*. [7] Shain et al. (2024). *Proceedings of the National Academy of Sciences.* [8] Kuribayashi et al. (2024). *Findings of NAACL*. [9] Hu et al. (2023). *Proceedings of ACL*. [10] Pan & Kehler (2025) *Proceedings of Cogsci*. [11] Rohde et al. (2011). *Cognition*.